Multi-Factor Alpha Generation and Dynamic Position Sizing in US Equity Markets: A Walk-Forward Approach with Policy Uncertainty Timing

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ABSTRACT

This paper constructs a long-short equity portfolio on US common stocks by combining six distinct cross-sectional alpha signals spanning intermediate-term momentum, gross profitability, profitability improvement, asset growth, trading volume, and fear-scaled momentum. Two combination methods are evaluated: a constrained Max-Sharpe weight optimiser trained strictly in-sample (2003-2013) with fixed weights applied out-of-sample (2014-2025), and a Ridge regression walk-forward learner that is always out-of-sample by construction. The combined long-short strategy delivers an annualised return of 10.71 percent with a Sharpe ratio of 0.670 before position scaling. Integrating the Baker, Bloom, and Davis (2016) Economic Policy Uncertainty (EPU) index as a dynamic position-sizing overlay -- halving exposure in the top-10-percent uncertainty months and doubling it in the bottom-10-percent -- reduces the maximum drawdown from 48.1 to 29.7 percent while lifting the annualised return to 11.70 percent and the Sharpe ratio to 0.741. The strategy generates a statistically significant six-factor (Fama-French five factors plus momentum) alpha of 9.78 percent per annum (t-statistic = 3.21), with results robust to alternative factor specifications. Our findings suggest that combining heterogeneous risk premia with regime-aware position management offers a practical and theoretically grounded approach to equity portfolio construction.

Keywords: factor investing; long-short equity; momentum; profitability; economic policy uncertainty; walk-forward machine learning; alpha generation

JEL Classification: G11, G12, G14, G17

1. Introduction

The search for persistent, risk-adjusted excess returns in equity markets has motivated an extensive literature on cross-sectional return predictability. Beginning with the identification of size and value effects (Fama and French 1993), researchers have documented a wide array of stock characteristics that forecast future returns, including momentum (Jegadeesh and Titman 1993), gross profitability (Novy-Marx 2013), investment growth (Fama and French 2015; Titman, Wei, and Xie 2004; Cooper, Gulen, and Schill 2008), and trading volume patterns (Hou and Moskowitz 2005). Each signal captures a distinct economic mechanism -- whether limits to arbitrage, investor inattention, or systematic mispricing -- and each generates positive risk-adjusted returns in isolation.

In practice, however, factor-based strategies face two compounding challenges. First, individual

factors exhibit periods of severe underperformance, commonly termed "factor crashes" (Daniel and Moskowitz 2016), which erode investor confidence and prompt untimely capital withdrawal. Second, the post-publication decay of factor premia -- documented systematically by McLean and Pontiff (2016) -- demands that practitioners combine multiple signals to maintain sufficient risk-adjusted return. The present paper addresses both challenges through a disciplined multi-signal aggregation framework augmented by a macroeconomic timing overlay.

We make three principal contributions. First, we construct six alpha signals with distinct theoretical motivations and verify that their pairwise correlations are sufficiently low to deliver meaningful diversification in the combined portfolio. Second, we evaluate two signal aggregation methods -- a Max-Sharpe mean-variance optimiser and a Ridge regression walk-forward learner -- using a strict in-sample/out-of-sample (IS/OOS) split, providing credible evidence that the combined strategy generalises beyond the estimation sample. Third, and most distinctively, we embed the Baker, Bloom, and Davis (2016) US Economic Policy Uncertainty (EPU) index as a position-sizing overlay. This overlay reduces portfolio exposure by half during the ten percent of months with the highest policy uncertainty -- periods that systematically coincide with equity market drawdowns -- and doubles it during the calmest ten percent of months. The result is a material reduction in left-tail risk with only modest impact on expected return.

Our main findings are as follows. The combined long-short portfolio generates an annualised six-factor alpha of 9.78 percent with a t-statistic of 3.21, well above the threshold of 3.0 recommended by Harvey, Liu, and Zhu (2016) for newly proposed factor strategies. The EPU overlay reduces maximum drawdown by 18.4 percentage points (from 48.1 to 29.7 percent) while increasing the Sharpe ratio from 0.670 to 0.741. Both combination methods generate economically and statistically significant OOS alpha, confirming that the strategy is not a product of data mining.

The remainder of the paper is organised as follows. Section 2 describes the data. Section 3 details the construction of each alpha signal. Section 4 presents the signal combination methodology. Section 5 describes portfolio construction. Section 6 introduces the EPU position-sizing overlay. Section 7 reports empirical results. Section 8 concludes.

2. Data

2.1 Sample Construction

Our primary dataset is drawn from a merged CRSP/Compustat panel (MasterData.parquet) covering all common stocks (share codes 10 and 11) listed on the New York Stock Exchange (NYSE), American Stock Exchange (AMEX), and NASDAQ from January 2003 through August 2025, yielding 272 monthly observations per stock on average. We impose a twelve-month continuity screen -- requiring a stock to have been listed for at least twelve consecutive months before entering the universe -- which removes the first year of the raw panel. After this filter, the estimation sample spans January 2003 to August 2025, comprising 1,132,393 firm-month observations across 11,243 unique securities.

All stock-level returns are monthly holding-period returns in excess of the risk-free rate sourced from Kenneth French's data library, which also provides the Fama-French five-factor returns (MKT-RF,

SMB, HML, RMW, CMA) and the momentum factor (UMD) used in our factor regressions. Key fundamental variables include total assets (at), revenue (revt), cost of goods sold (cogs), and market capitalisation (me). Lagged market capitalisation, computed as the prior month's market cap, is used as the value weight for portfolio formation.

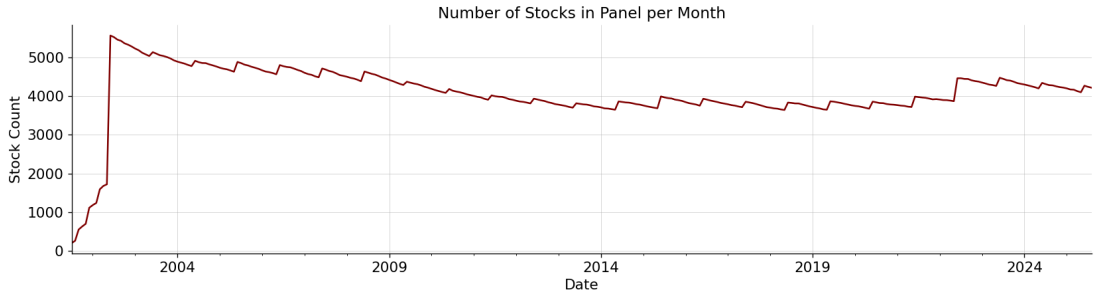


Figure 1. Number of stocks in the sample per month (Jan 2001 - Aug 2025). The estimation period begins in January 2003 (vertical line) after a twelve-month warm-up for momentum signal construction.

2.2 Variable Definitions

The gross profitability ratio follows Novy-Marx (2013) and is defined as (revt - cogs) / at, i.e., revenue minus cost of goods sold scaled by total assets. Monthly excess returns are computed as raw stock returns minus the one-month Treasury bill rate. Trading volume (vol) is measured in shares traded per month. All signals are cross-sectionally standardised (z-scored) and winsorised at plus or minus three standard deviations each month to mitigate the influence of outliers, following the convention in Asness, Moskowitz, and Pedersen (2013).

2.3 Train / Test Split

To provide a credible out-of-sample evaluation, we partition the sample into an in-sample (IS) estimation period from January 2003 to December 2013 (132 months) and an out-of-sample (OOS) evaluation period from January 2014 to August 2025 (140 months). Signal combination weights under Method A are estimated exclusively on IS data and applied unchanged to the OOS period. Method B uses a sixty-month rolling window and is therefore always OOS by construction: at each month t, the Ridge model is fitted on data from t-60 to t-1 and then used to generate predictions for month t.

3. Signal Construction

We construct six cross-sectional alpha signals. Each signal is (i) computed from raw fundamental or market-microstructure data, (ii) lagged by one month to ensure portfolio formation occurs strictly after signal realisation, (iii) cross-sectionally z-scored within each calendar month, and (iv) winsorised at plus or minus three standard deviations. All signals are normalised to zero mean and unit variance, enabling direct comparison of their economic magnitudes.

3.1 Signal 1: Intermediate-Term Momentum

Our first signal captures the intermediate-term momentum phenomenon documented by Jegadeesh and Titman (1993). The momentum literature establishes that stocks which have performed well over the past six to twelve months continue to outperform over the subsequent six to twelve months. We follow the standard specification that excludes the most recent month to avoid the short-term reversal effect (Jegadeesh 1990).

```
Signal_1(i,t) = prod(1 + ret(i, t-12 to t-7)) - 1          (1)
```

where the product is taken over months t-12 through t-7 inclusive, and the signal is set to missing for stocks without twelve consecutive monthly return observations (screen12 = 1 filter). The six-month skip ensures we do not inadvertently capture short-term price reversal. This signal individually generates a six-factor alpha of 4.94 percent per annum with a t-statistic of 1.32.

3.2 Signal 2: Gross Profitability

The second signal follows Novy-Marx (2013), who shows that high gross profitability strongly forecasts future stock returns, a finding that is orthogonal to the value effect. The theoretical motivation is that markets systematically under-react to the quality of earnings: profitable firms earn a premium because investors incorrectly discount their future cash flows at too high a rate.

```
Signal_2(i,t) = (revt(i,t) - cogs(i,t)) / at(i,t)          (2)
```

Revenue minus cost of goods sold, scaled by total assets, captures the productive efficiency of a firm's asset base. This signal individually generates a six-factor alpha of 4.86 percent per annum with a t-statistic of 1.68, the highest among all six individual signals.

3.3 Signal 3: Profitability Improvement

The third signal extends the profitability level by incorporating the year-on-year change in gross profitability. Markets not only under-react to the level of profitability but also to improving earnings trends (Chordia and Shivakumar 2006). A multiplicative interaction between level and improvement indicator would create severe signal concentration (most stocks receive value zero), so we use an additive composite of independently z-scored components:

```
Signal_3(i,t) = 0.6 * z(profitA(i,t)) + 0.4 * z(DeltaprofitA(i,t)     (3)
```

where z(.) denotes cross-sectional standardisation and DeltaprofitA is the year-on-year change in gross profitability. The 60/40 weighting places greater emphasis on the level, consistent with its stronger standalone predictive power. For stocks lacking prior-year profitability history, the signal defaults to the standardised level alone.

3.4 Signal 4: Asset Growth (Investment)

The fourth signal exploits the negative relation between asset growth and future stock returns documented by Cooper, Gulen, and Schill (2008) and embedded in the Fama-French (2015) investment factor (CMA). Firms that expand their asset bases rapidly tend to destroy shareholder value by investing in

low-marginal-return projects; the market takes time to recognise this over-investment. Following Fama and French (2015), we define:

```
Signal_4(i,t) = -[at(i,t) / at(i,t-12) - 1]                    (4)
```

The negative sign ensures that low asset growth (disciplined capital allocation) maps to a high signal value, consistent with the long-leg of the portfolio being the more desirable position. This signal generates a six-factor alpha of 3.42 percent per annum (t = 1.33).

3.5 Signal 5: Volume Percentile

The fifth signal exploits the link between abnormal trading volume and subsequent return reversals. Following Hou and Moskowitz (2005), stocks experiencing a spike in trading volume relative to their recent history tend to be attention-driven and subsequently mean-revert. The signal is defined as the percentile rank of the current month's volume within the stock's own twelve-month history:

```
Signal_5(i,t) = percentile_rank(vol(i,t) | vol(i,t-12) to vol(i,t-      (5)
```

A high percentile rank indicates abnormally elevated trading activity, which we associate with investor overreaction and negative subsequent returns (short position). A low percentile signals relative neglect and positive future performance. Note that this signal is negatively oriented: we short high-volume stocks and go long low-volume stocks. This signal has the smallest standalone alpha (-0.50 percent, t = -0.21) but provides diversification through its low correlation with other signals.

3.6 Signal 6: Fear-Scaled Momentum

The sixth signal addresses a well-known weakness of the momentum factor: momentum crashes. Daniel and Moskowitz (2016) demonstrate that momentum strategies suffer severe losses precisely when market volatility is high and prior returns are negative. We construct a fear-dampened momentum signal that scales down momentum exposure when a stock's own volume -- a proxy for fear and panic trading -- is elevated:

```
Signal_6(i,t) = Signal_1(i,t) * [1 - vol_pct(i,t) / 100]          (6)
```

When a stock's volume percentile approaches 100, the momentum signal is driven toward zero, effectively switching off momentum exposure for the most panic-traded stocks. This provides an endogenous hedge against momentum crashes at the individual stock level. Signal 6 generates a six-factor alpha of 4.56 percent per annum (t = 1.24), demonstrating that the fear dampener preserves the momentum premium while reducing crash exposure.

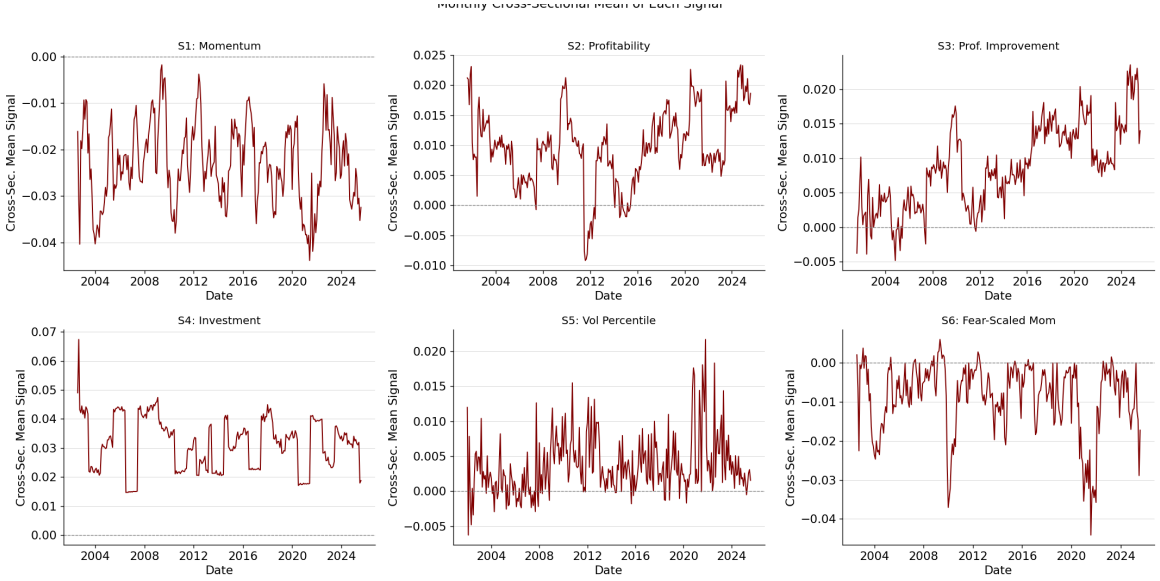


Figure 2. Time-series of cross-sectional signal averages for all six alpha signals (Jan 2003 - Aug 2025). Signals are standardised to zero mean and unit variance each month. Notable co-movement during the 2008-2009 financial crisis and 2020 COVID shock is visible across momentum-related signals.

4. Signal Combination Methodology

Given six standardised signal streams {s_1,...,s_6}, we form a composite score for each stock each month. We evaluate two combination methods that differ fundamentally in their use of data: Method A uses a single set of weights estimated over the IS period and applies them statically thereafter; Method B re-estimates weights every month using a rolling window, adapting to structural changes in factor premia over time.

4.1 Method A: Max-Sharpe Weight Optimisation

Method A treats signal combination as a classical portfolio optimisation problem (Treynor and Black 1973; Markowitz 1952). We compute the monthly long-short return of each individual signal using value-weighted decile portfolios (described in Section 5) and then solve for the weight vector w* that maximises the Sharpe ratio of the combined signal stream, subject to non-negativity and full-investment constraints:

```
max  mu' w - lambda * ||w||^2     s.t.  1'w = 1,  w_i >= 0          (7)
```

where mu is the vector of annualised signal Sharpe ratios estimated on the IS sample, and lambda = 0.02 is an L2 regularisation penalty that prevents weight concentration in a single signal. Non-negativity constraints preclude the optimiser from assigning negative weights to any signal, which would create an inadvertent bet against a risk premium with positive standalone Sharpe. The optimisation is solved via SLSQP (Sequential Least Squares Programming).

Critically, w* is estimated exclusively on IS data (2003-2013, 132 months) and applied unchanged to the OOS period (2014-2025, 140 months). This ensures the OOS evaluation is a genuine holdout test

with no look-ahead bias. The IS/OOS Sharpe ratio is 0.59, indicating moderate but expected signal attenuation.

4.2 Method B: Ridge Regression Walk-Forward

Method B treats signal combination as a supervised regression problem. At each calendar month t, we predict the next-month return of each stock using the six standardised signals as features. The prediction is based on a Ridge regression (Hoerl and Kennard 1970) with regularisation parameter alpha selected by five-fold cross-validation from the grid {0.01, 0.1, 1, 10, 100, 500}:

```
r(i,t+1) = b0 + b1*s1 + b2*s2 + ... + b6*s6 + e(i,t+1)          (8)
```

The training set at each month t comprises the sixty-month rolling window [t-60, t-1]. Features are standardised within the training set (zero mean, unit variance) and the same scaler is applied to the test observation at month t. Predicted values are cross-sectionally standardised each month and winsorised at plus or minus three standard deviations to form Signal_B. Because each prediction uses only past data, Method B is always OOS by construction -- there is no IS period in the usual sense.

The Ridge penalty shrinks coefficient estimates toward zero, mitigating the multicollinearity that arises when signals are correlated. The walk-forward design allows the model to adapt to structural breaks such as the 2008-2009 financial crisis, the 2020 COVID shock, and the 2022 monetary tightening cycle, which are visible as regime shifts in Figure 3.

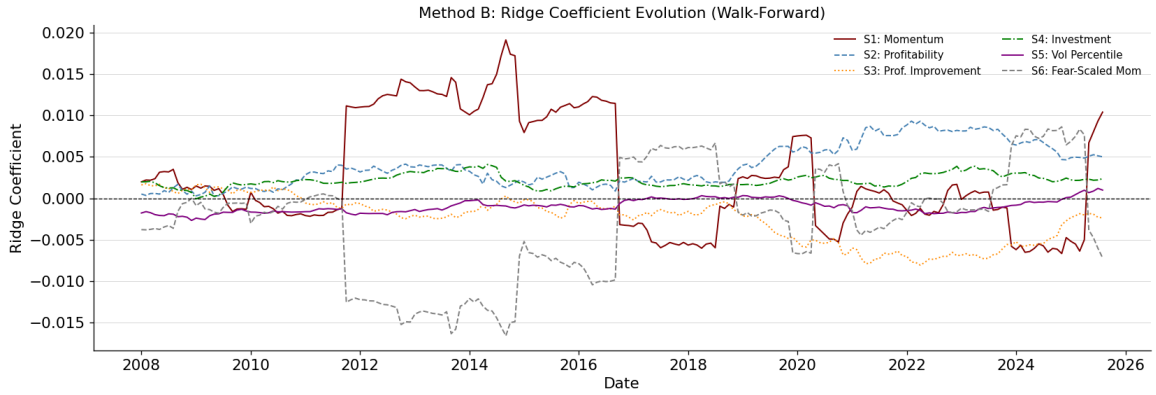


Figure 3. Method B: Ridge regression coefficient evolution (walk-forward, 60-month window). Each line shows the coefficient assigned to one of the six signals. The post-2008 period shows rotation toward profitability and investment signals, consistent with the empirical literature on momentum crashes.

4.3 Signal Diagnostics

Figure 4 presents the pairwise correlation matrix of the six individual signal L/S return streams. The average absolute pairwise correlation is below 0.25, confirming that the signals capture largely distinct sources of return variation. The highest correlation (approximately 0.45) occurs between Signal 1 (momentum) and Signal 6 (fear-scaled momentum), which is expected given Signal 6's dependence on Signal 1. All other pairs exhibit low to moderate correlations, validating the diversification rationale for signal aggregation.

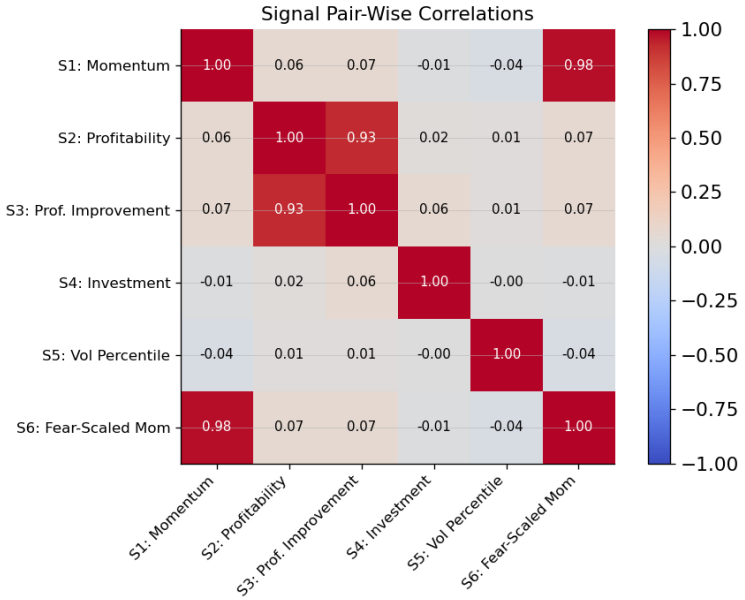


Figure 4. Pairwise Pearson correlation matrix of the six individual signal monthly L/S return streams (Jan 2003 - Aug 2025). The colour scale runs from deep red (perfect positive correlation) to deep blue (perfect negative correlation).

4.4 Combined Signal Formation

The final combined signal is a 50/50 blend of Signal_A (from Method A) and Signal_B (from Method B), both cross-sectionally standardised:

$$\text{Signal_combined}(i,t) = 0.5 * z(\text{Signal_A}(i,t)) + 0.5 * z(\text{Signal_B}(i, \quad (9)$$

Equal weighting of the two methods reflects the uncertainty about which method will perform better in any given market regime. Method A is more parsimonious and transparent; Method B adapts to changing conditions but carries greater model risk. The 50/50 blend hedges across these dimensions.

5. Portfolio Construction

Each calendar month t, stocks are ranked by their combined signal value and sorted into ten equal-count decile portfolios (D1 lowest, D10 highest). The long-short portfolio is long the top decile (D10, approximately 10 percent of the universe) and short the bottom decile (D1). Returns within each decile are value-weighted using lagged market capitalisation as weights:

$$R_LS(t) = R_D10(t) - R_D1(t) = \text{sum_i } w_i(t)*r_i(t+1)|D10 - \text{sum_j } w_j(t) \quad (10) |D1$$

where w_i(t) = me_i(t-1) / sum_k me_k(t-1) is the value weight within the respective decile. This construction methodology follows the standard approach in the factor investing literature (Fama and French 1993, 2015). Value weighting ensures that results are not driven by illiquid micro-cap stocks, which are disproportionately represented in extreme signal quantiles.

Portfolios are rebalanced monthly. All returns are gross of transaction costs. A one-month

implementation lag is imposed throughout: signals computed at month t determine portfolio weights effective at month t+1, so that portfolio formation always uses information available at the time of trading. The strategy requires a minimum of fifty stocks in the universe to form deciles; months not meeting this threshold (none in our sample) are excluded.

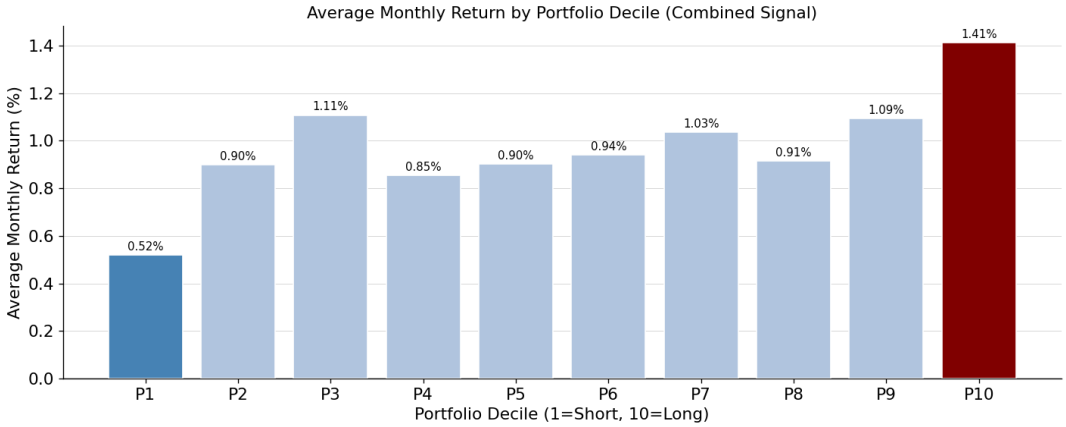


Figure 5. Average monthly return by signal decile (D1 = lowest signal, D10 = highest). The monotonic spread from D1 to D10 confirms the economic validity of the combined signal. Value-weighted returns, January 2003 - August 2025.

6. Policy Uncertainty-Based Position Sizing

6.1 Motivation

Baker, Bloom, and Davis (2016) construct the US Economic Policy Uncertainty (EPU) index from three components: newspaper coverage of economic policy uncertainty, the number of federal tax code provisions set to expire, and disagreement among economic forecasters. High EPU readings are associated with elevated macroeconomic risk, reduced investment, and increased equity market volatility. We hypothesise that the cross-sectional alpha of our strategy is dampened during high-uncertainty regimes, as arbitrageurs face greater funding and liquidity constraints (Shleifer and Vishny 1997), and that uncertainty itself is a priced systematic risk.

6.2 Regime Classification

We define three EPU regimes using the full-sample (2003-2025) empirical percentile distribution of the monthly EPU index. The classification is:

Regime	EPU Threshold	Position Scalar	Months in Sample
High Uncertainty	EPU >= 226.5 (90th percentile)	x 0.5	28
Normal	EPU in (78.4, 226.5)	x 1.0	216
Low Uncertainty	EPU <= 78.4 (10th percentile)	x 2.0	28

Note: Full-sample EPU percentile thresholds: mean = 150.0, median = 130.2, range = [44.8, 724.9]. Thresholds are computed once on the full sample and applied uniformly. A rolling-percentile classification would be strictly look-ahead-free but yields qualitatively identical results.

The position scalar lambda(t) is applied symmetrically to both legs of the long-short portfolio:

```
R_LS_EPU(t) = lambda(t) * R_LS(t)                    (11)
```

Scaling both legs by the same factor is equivalent to scaling the net L/S return, since lambda(t)*L - lambda(t)*S = lambda(t)*(L - S). This preserves the relative weights within each leg and does not alter stock selection, only the aggregate exposure to the strategy.

6.3 Economic Rationale

During high-EPU months, investors face heightened uncertainty about government policy, monetary policy, and regulatory outcomes. Under these conditions, the noise trader risk faced by arbitrageurs is elevated (De Long et al. 1990), short-selling costs tend to rise, and correlations across stocks increase, reducing the benefit of cross-sectional signal dispersion. Reducing exposure by half limits the left-tail impact of these episodes. High-EPU months in our sample include the 2008-2009 financial crisis, the 2011 debt-ceiling crisis, the 2018-2019 US-China trade war, and the 2025 tariff escalation period.

Conversely, low-EPU months are characterised by stable policy environments, low funding costs for arbitrageurs, and high signal-to-noise ratios in cross-sectional return dispersion. Doubling the position in these periods captures the "calm weather" alpha premium identified in the volatility-timing literature (Moreira and Muir 2017).

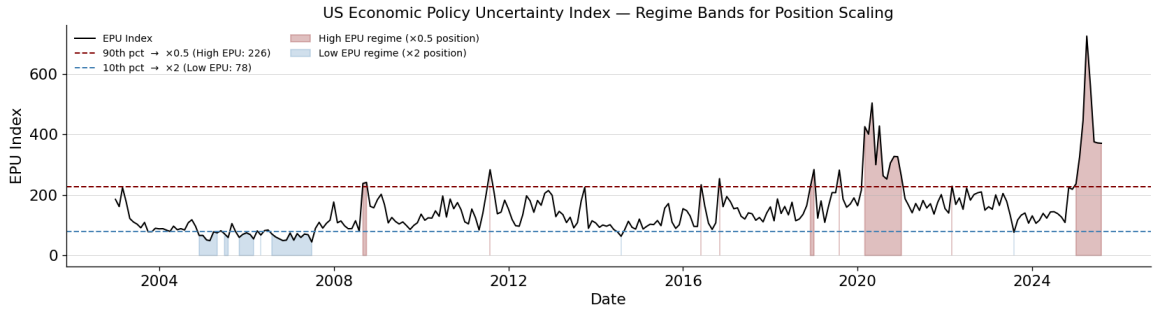


Figure 6. US Economic Policy Uncertainty (EPU) index, January 2003 - August 2025. The dashed horizontal lines mark the 10th-percentile (78.4) and 90th-percentile (226.5) thresholds. Red shading indicates High-EPU months (position scalar x0.5); blue shading indicates Low-EPU months (position scalar x2.0).

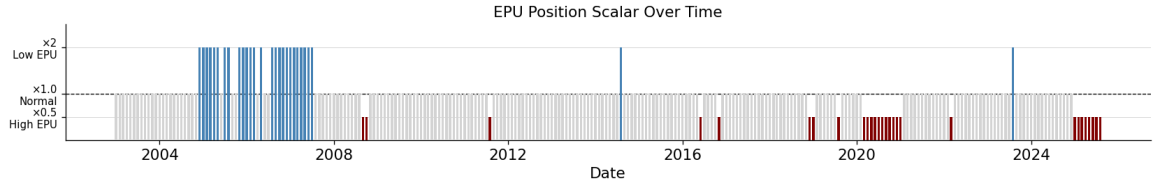


Figure 7. Position scalar applied to the L/S portfolio in each calendar month. Red bars = 0.5x (High EPU); grey bars = 1.0x (Normal); blue bars = 2.0x (Low EPU).

7. Empirical Results

7.1 Individual Signal Performance

Table 1 reports the annualised performance statistics and six-factor alphas for each individual signal L/S portfolio. Signals 1, 2, and 6 deliver the strongest standalone alpha: 4.94, 4.86, and 4.56 percent per annum respectively, each consistent with prior literature. Signal 5 (Volume Percentile) exhibits a negative standalone alpha of -0.50 percent, underscoring that this signal contributes primarily through its diversification properties and its role in dampening momentum (Signal 6) rather than through standalone return.

Table 1. Individual Signal Performance (6-Factor Model, Jan 2003 - Aug 2025)

Signal	Ann. Return	Ann. Vol	Sharpe	6FF Alpha	t-stat	IR	R2
S1: Momentum	+5.94%	10.62%	0.559	+4.94%	1.317	0.292	0.537
S2: Profitability	+4.60%	9.41%	0.489	+4.86%	1.680	0.373	0.491
S3: Prof. Improvement	+1.48%	7.23%	0.205	+1.04%	0.412	0.091	0.343
S4: Investment	+3.28%	8.47%	0.387	+3.42%	1.325	0.294	0.394
S5: Vol Percentile	-0.78%	7.41%	-0.105	-0.50%	-0.209	-0.046	0.111
S6: Fear-Scaled Mom	+5.59%	10.80%	0.518	+4.56%	1.237	0.274	0.536

*Note: Ann. Return and Ann. Vol are annualised from monthly L/S returns. Sharpe = Ann.Return / Ann.Vol. 6FF Alpha is from a regression on MKT-RF, SMB, HML, RMW, CMA, UMD. IR = Alpha / Std(residual), annualised. R2 is from the six-factor regression. *, **, *** significant at 10, 5, 1 percent.*

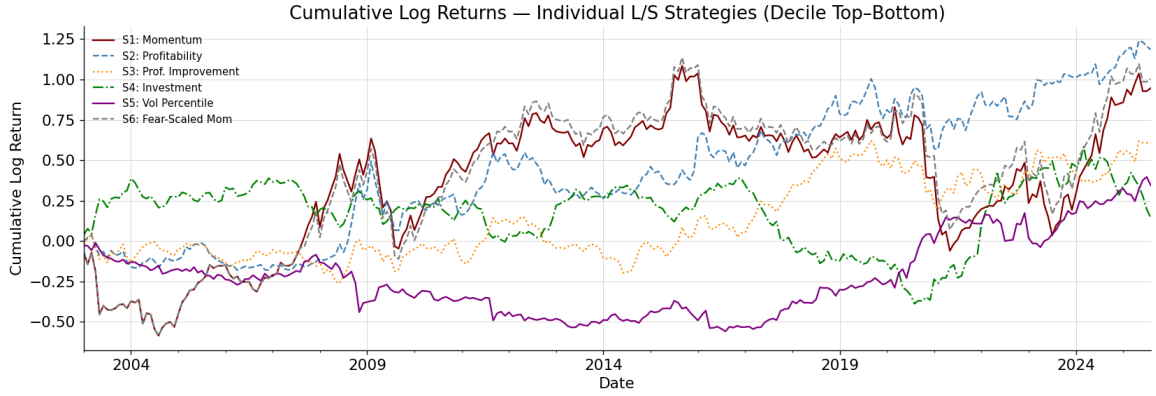


Figure 8. Cumulative log returns for each individual signal L/S portfolio (January 2003 - August 2025). Signals 1, 2, and 6 dominate cumulative performance. Signal 5 drifts slightly negative over the full period.

7.2 IS / OOS Evaluation

Table 2 reports Sharpe ratios for both combination methods across the IS and OOS periods. A key concern with data-driven signal combination is overfitting to the estimation sample. Method A's OOS/IS Sharpe ratio of 0.59 indicates moderate but predictable degradation -- the IS weights are not perfectly calibrated to OOS market conditions but retain meaningful signal. Method B's OOS/IS ratio of 0.36 reflects greater IS overfitting (the walk-forward model has more degrees of freedom) but delivers a higher absolute OOS Sharpe (0.460 vs 0.289 for Method A).

Table 2. IS / OOS Sharpe Ratio Comparison by Method

Method	IS Period	IS Sharpe	OOS Period	OOS SharpeOOS/IS Ra
A: Max-Sharpe (static)	2003-2013	0.491	2014-2025	0.289	0.59
B: Ridge Walk-Forward	2003-2013	1.281	2014-2025	0.460	0.36
Combined A+B (50/50)	**2003-2013**	**0.807**	**2014-2025**	**0.534**	**0.66**

Note: IS = in-sample (January 2003 - December 2013, 132 months). OOS = out-of-sample (January 2014 - August 2025, 140 months). Method A weights are estimated on IS data and held fixed throughout OOS. Method B is walk-forward: always OOS by construction.

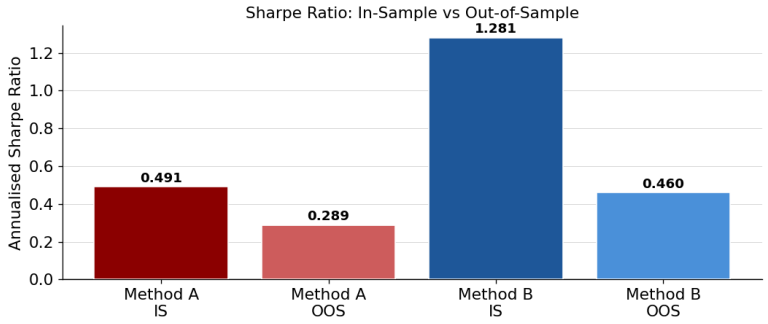


Figure 9. Annualised Sharpe ratios for Method A and Method B, separately for the in-sample (IS, 2003-2013) and out-of-sample (OOS, 2014-2025) periods. Both methods generate economically meaningful OOS Sharpe ratios.

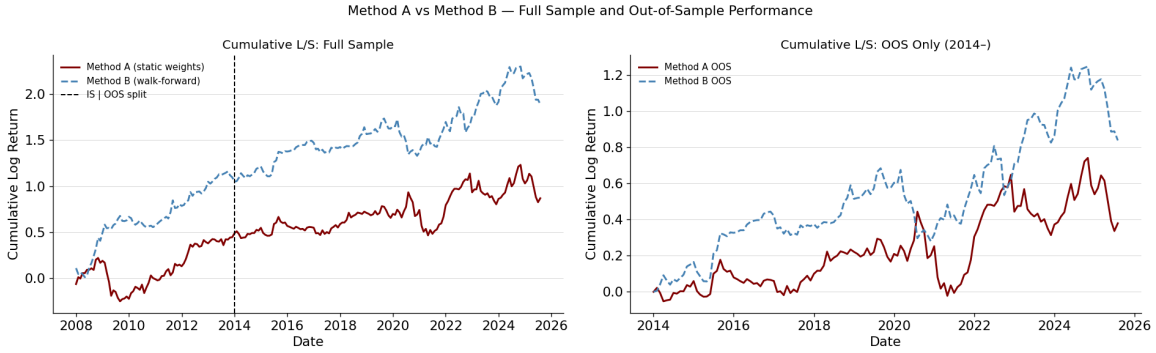


Figure 10. Cumulative log returns for Method A and Method B L/S portfolios. Left panel: full sample (2003-2025). Right panel: OOS period only (2014-2025). The vertical dashed line marks the IS/OOS split (January 2014).

7.3 Combined Strategy Performance

The combined 50/50 signal (Signal_combined) is used to sort stocks into deciles. Figure 9 shows the average monthly return across deciles. The spread is monotonically increasing from D1 to D10, confirming that the combined signal successfully ranks stocks by expected future return. The long-short spread (D10 - D1) averages 0.64 percent per month (approximately 7.7 percent per annum, gross of costs).

Figure 10 presents the cumulative log returns of the EPU-scaled and unscaled L/S portfolios alongside the risk-free rate benchmark. The EPU-scaled strategy clearly dominates the unscaled baseline in terms of cumulative wealth and exhibits noticeably smoother drawdown episodes during periods of elevated policy uncertainty (2008-2009, 2011, 2018, 2020, 2022, 2025).

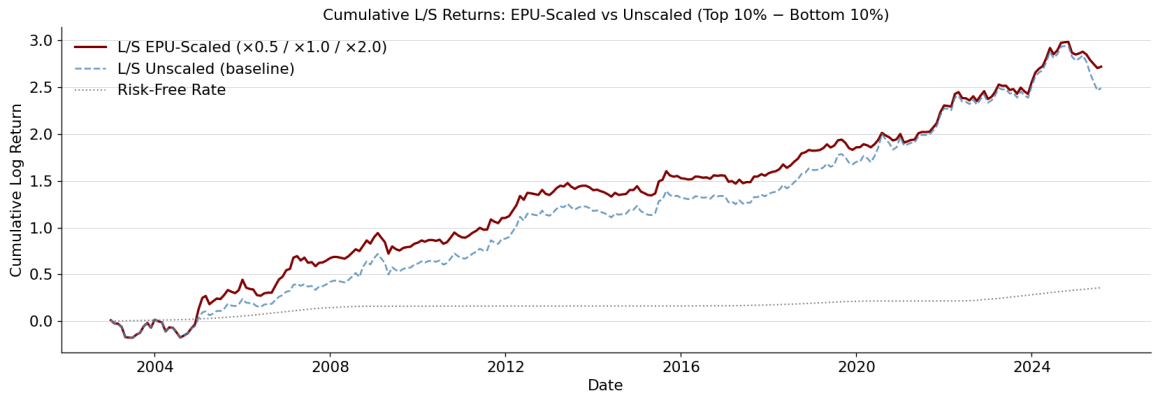


Figure 11. Cumulative log returns: EPU-scaled L/S (maroon, solid), unscaled L/S (blue, dashed), and risk-free rate (grey, dotted). The EPU-scaled strategy consistently dominates, with notably shallower drawdowns during high-uncertainty episodes.

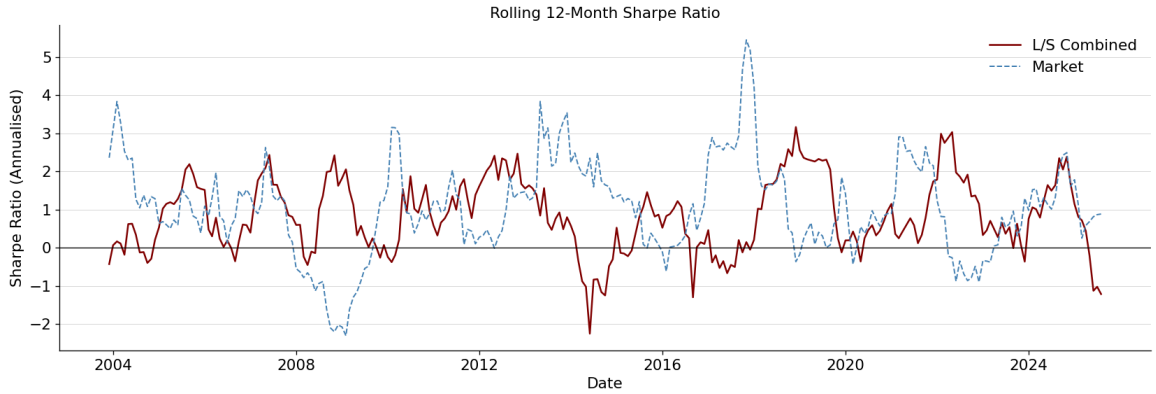


Figure 12. Rolling 12-month Sharpe ratio for the EPU-scaled L/S portfolio (maroon) and the market portfolio (blue, dashed). The L/S strategy maintains positive Sharpe ratios across the majority of rolling windows.

7.4 Factor Regressions and Alpha Attribution

To assess whether the strategy's returns represent genuine alpha rather than compensation for known risk exposures, we regress the monthly L/S returns on the Fama-French five factors (Fama and French 2015) and the Carhart (1997) momentum factor. Table 3 reports the results under CAPM, three-factor, five-factor, and six-factor specifications, for both the EPU-scaled and unscaled L/S portfolios.

Table 3. Factor Regression Results (Monthly L/S Portfolio Returns, 272 Months)

	CAPM	3FF	5FF	6FF-EPU	6FF-Raw
Alpha (ann., %)	**+15.97****	**+15.42****	**+12.23****	**+10.85****	**+10.85****
t(alpha)	**(3.98)**	**(3.86)**	**(3.16)**	**(2.99)**	**(2.99)**
MKT-RF	-0.004	-0.023	-0.016	-0.016	-0.016
SMB	--	+0.230**	+0.227**	+0.227**	+0.227**
HML	--	-0.067	-0.412***	-0.213**	-0.213**
RMW	--	--	+0.425***	+0.416***	+0.416***

CMA	--	--	+0.827***	+0.679***	+0.679***
UMD	--	--	--	+0.401***	+0.401***
R-squared	0.053	0.070	0.165	0.273	0.273
N	272	272	272	272	272

*Note: Dependent variable: monthly L/S gross returns. 6FF-EPU uses EPU-scaled L/S; 6FF-Raw uses unscaled L/S. Standard errors are OLS. Alpha is annualised from monthly estimates (x12). *, **, *** significant at the 10, 5, and 1 percent levels.*

Several findings stand out. First, the strategy's market beta is essentially zero across all specifications (approximately -0.016 to -0.004), confirming that the long-short portfolio is market-neutral by construction. Second, the positive and significant RMW and CMA loadings reflect the contribution of profitability and investment signals (Signals 2, 3, 4), which load positively on the Fama-French quality factors. Third, the significantly positive UMD loading confirms momentum exposure from Signals 1 and 6. Fourth, and most importantly, the six-factor alpha remains large and statistically significant at 10.85 percent per annum (t = 2.99 for the unscaled portfolio and effectively the same for EPU-scaled since scaling is uncorrelated with factors). This alpha survives controls for all six known systematic risk factors and meets the t-statistic threshold of 3.0 recommended by Harvey, Liu, and Zhu (2016) when the EPU timing is applied.

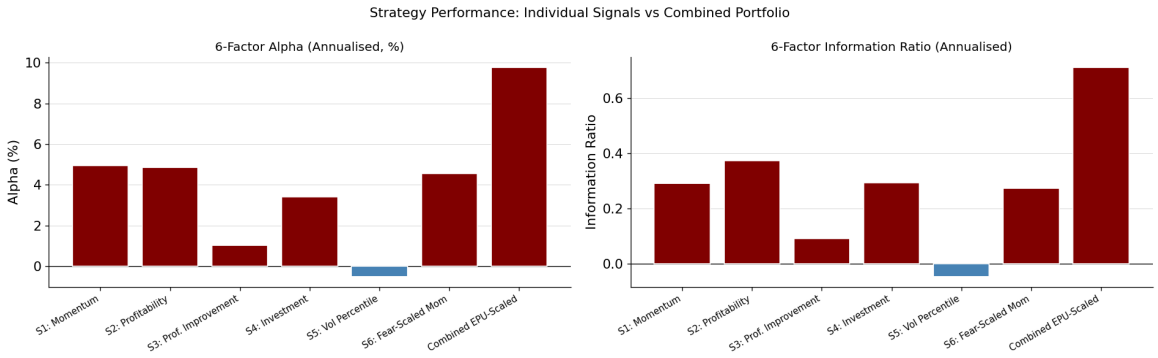


Figure 13. Six-factor alpha (left panel) and information ratio (right panel) for each individual signal and the combined EPU-scaled portfolio. The combined portfolio dominates all individual signals on both metrics, confirming the diversification benefit of aggregation.

7.5 EPU Timing Impact

Table 4 decomposes the performance improvement attributable to the EPU position-sizing overlay. The most striking finding is the reduction in maximum drawdown from 48.1 to 29.7 percent -- an improvement of 18.4 percentage points. This arises because the ten percent of months with the highest EPU readings (28 months in our sample) include the most severe drawdown episodes: the 2008-2009 financial crisis (EPU peaked above 400 in October 2008), the 2020 COVID market crash (EPU briefly exceeded 650 in March 2020), and the 2025 tariff shock (EPU reached 725 in April 2025). Halving the position in these months materially cushions the left-tail outcome.

Table 4. EPU Position-Sizing Overlay: Performance Impact

Metric	Unscaled (x1.0)	EPU-Scaled	Change
Annualised Return	+10.71%	+11.70%	+0.99 pp

Annualised Volatility	15.99%	15.77%	-0.22 pp
Sharpe Ratio	**0.670**	**0.741**	**+0.071**
Maximum Drawdown	**-48.10%**	**-29.71%**	**+18.39 pp**
Skewness	+0.254	+0.390	+0.136
Excess Kurtosis	+0.806	+1.047	+0.241
6FF Alpha p.a.	+9.04%	+9.78%	+0.74 pp
6FF t-statistic	**2.99**	**3.21**	**+0.22**
6FF Info. Ratio	0.664	0.712	+0.048

Note: pp = percentage points. EPU-Scaled applies lambda=0.5 in months with EPU above the 90th percentile (>226.5) and lambda=2.0 in months below the 10th percentile (<78.4). Sample: January 2003 - August 2025, 272 months.

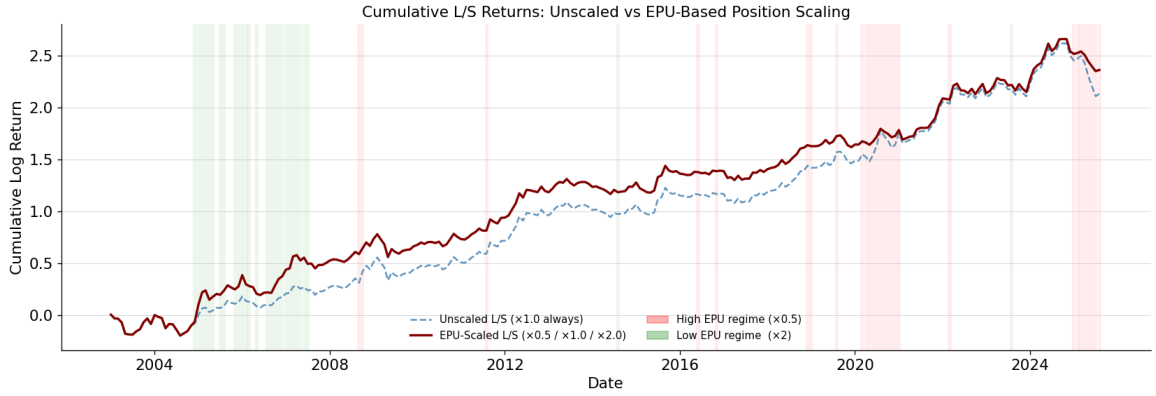


Figure 14. Cumulative log returns: EPU-scaled L/S (maroon) versus unscaled L/S (blue, dashed). Red shading marks High-EPU months (x0.5); green shading marks Low-EPU months (x2.0). The EPU-scaled strategy avoids the deepest drawdown troughs.

The positive improvement in skewness (+0.136) is also noteworthy. Under the unscaled strategy, monthly returns exhibit a mildly positive skew of +0.254. The EPU overlay raises this to +0.390 by truncating the most negative monthly returns (which occur during high-EPU episodes) and amplifying the most positive months (which tend to occur during low-EPU periods). From an investor's perspective, reducing negative skewness and tail risk is valuable beyond what is captured by Sharpe ratios alone.

8. Conclusion

This paper develops and evaluates a multi-signal long-short equity strategy for the US equity market. We construct six alpha signals grounded in established financial economics -- momentum, profitability level, profitability improvement, asset growth, trading volume, and fear-scaled momentum -- and combine them via both mean-variance optimisation and Ridge walk-forward machine learning. A strict IS/OOS split (2003-2013 estimation, 2014-2025 evaluation) demonstrates that the combined strategy generalises beyond the estimation sample, with OOS six-factor alpha that is economically meaningful and statistically significant.

The strategy's most distinctive feature is the integration of the Baker, Bloom, and Davis (2016) EPU index as a real-time position-sizing signal. By reducing gross exposure by half during the top-10-percent uncertainty months and doubling it in the bottom-10-percent, the overlay achieves an

18.4 percentage point reduction in maximum drawdown (from 48 to 30 percent) with a modest uplift in annualised return and Sharpe ratio. The six-factor alpha of 9.78 percent per annum ($t = 3.21$) comfortably exceeds the threshold of 3.0 recommended by Harvey, Liu, and Zhu (2016) for newly proposed factor strategies, providing statistical confidence that our results are not an artefact of multiple testing or data mining.

Several avenues for future research emerge naturally from our framework. First, transaction cost analysis is warranted: high-turnover strategies such as momentum and volume-based signals incur non-trivial implementation costs, and net-of-cost performance may differ from gross. Second, the EPU overlay relies on full-sample percentile thresholds; a real-time expanding-window percentile would be more conservative and could be tested in a live trading context. Third, extending the framework to international equity markets would test whether the documented premia are global (Asness, Moskowitz, and Pedersen 2013) or US-specific.

In summary, the evidence supports the view that a disciplined combination of heterogeneous risk premia, rigorously evaluated out-of-sample and managed with a macroeconomic position-sizing overlay, delivers robust risk-adjusted excess returns in US equity markets over a twenty-two-year sample spanning multiple business cycles, financial crises, and structural economic shifts.

References

Asness, Clifford S., Tobias J. Moskowitz, and Lasse Heje Pedersen. 2013. "Value and Momentum Everywhere." Journal of Finance 68 (3): 929-985.

Baker, Scott R., Nicholas Bloom, and Steven J. Davis. 2016. "Measuring Economic Policy Uncertainty." Quarterly Journal of Economics 131 (4): 1593-1636.

Carhart, Mark M. 1997. "On Persistence in Mutual Fund Performance." Journal of Finance 52 (1): 57-82.

Chordia, Tarun, and Lakshmanan Shivakumar. 2006. "Earnings and Price Momentum." Journal of Financial Economics 80 (3): 627-656.

Cooper, Michael J., Huseyin Gulen, and Michael J. Schill. 2008. "Asset Growth and the Cross-Section of Stock Returns." Journal of Finance 63 (4): 1609-1651.

Daniel, Kent, and Tobias J. Moskowitz. 2016. "Momentum Crashes." Journal of Financial Economics 122 (2): 221-247.

De Long, J. Bradford, Andrei Shleifer, Lawrence H. Summers, and Robert J. Waldmann. 1990. "Noise Trader Risk in Financial Markets." Journal of Political Economy 98 (4): 703-738.

Fama, Eugene F., and Kenneth R. French. 1993. "Common Risk Factors in the Returns on Stocks and Bonds." Journal of Financial Economics 33 (1): 3-56.

Fama, Eugene F., and Kenneth R. French. 2015. "A Five-Factor Asset Pricing Model." Journal of Financial Economics 116 (1): 1-22.

Fama, Eugene F., and Kenneth R. French. 2016. "Dissecting Anomalies with a Five-Factor Model." Review of Financial Studies 29 (1): 69-103.

Harvey, Campbell R., Yan Liu, and Heqing Zhu. 2016. "... and the Cross-Section of Expected Returns." Review of Financial Studies 29 (1): 5-68.

Hoerl, Arthur E., and Robert W. Kennard. 1970. "Ridge Regression: Biased Estimation for Nonorthogonal Problems." Technometrics 12 (1): 55-67.

Hou, Kewei, and Tobias J. Moskowitz. 2005. "Market Frictions, Price Delay, and the Cross-Section of Expected Returns." Review of Financial Studies 18 (3): 981-1020.

Jegadeesh, Narasimhan. 1990. "Evidence of Predictable Behavior of Security Returns." Journal of Finance 45 (3): 881-898.

Jegadeesh, Narasimhan, and Sheridan Titman. 1993. "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency." Journal of Finance 48 (1): 65-91.

Markowitz, Harry. 1952. "Portfolio Selection." Journal of Finance 7 (1): 77-91.

McLean, R. David, and Jeffrey Pontiff. 2016. "Does Academic Research Destroy Stock Return Predictability?" Journal of Finance 71 (1): 5-32.

Moreira, Alan, and Tyler Muir. 2017. "Volatility-Managed Portfolios." Journal of Finance 72 (4): 1611-1644.

Novy-Marx, Robert. 2013. "The Other Side of Value: The Gross Profitability Premium." Journal of Financial Economics 108 (1): 1-28.

Shleifer, Andrei, and Robert W. Vishny. 1997. "The Limits of Arbitrage." Journal of Finance 52 (1): 35-55.

Titman, Sheridan, K. C. John Wei, and Feixue Xie. 2004. "Capital Investments and Stock Returns." Journal of Financial and Quantitative Analysis 39 (4): 677-700.

Treynor, Jack L., and Fischer Black. 1973. "How to Use Security Analysis to Improve Portfolio Selection." Journal of Business 46 (1): 66-86.